ASCAT-DESTEKLI TIGGE RUZGAR HATA DUZELTMESI

Thiessen poligonlari ve XGBoost ile pilot dogrulama

Ozet

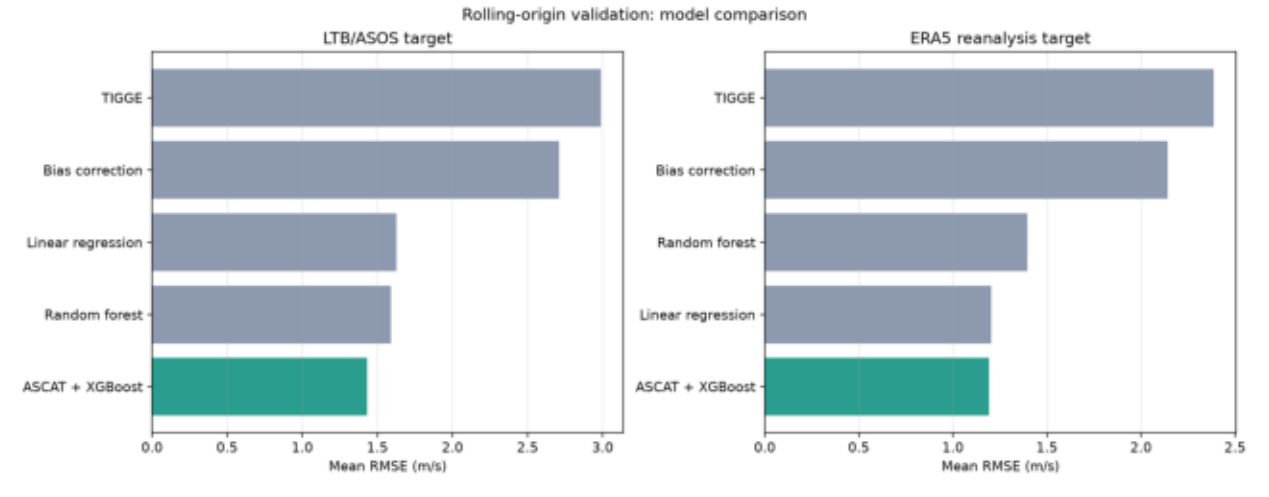
Bu rapor, 12 saatlik TIGGE ruzgar tahminlerini tahmin baslangicindaki ASCAT gozlemleriyle duz

Ana bulgu

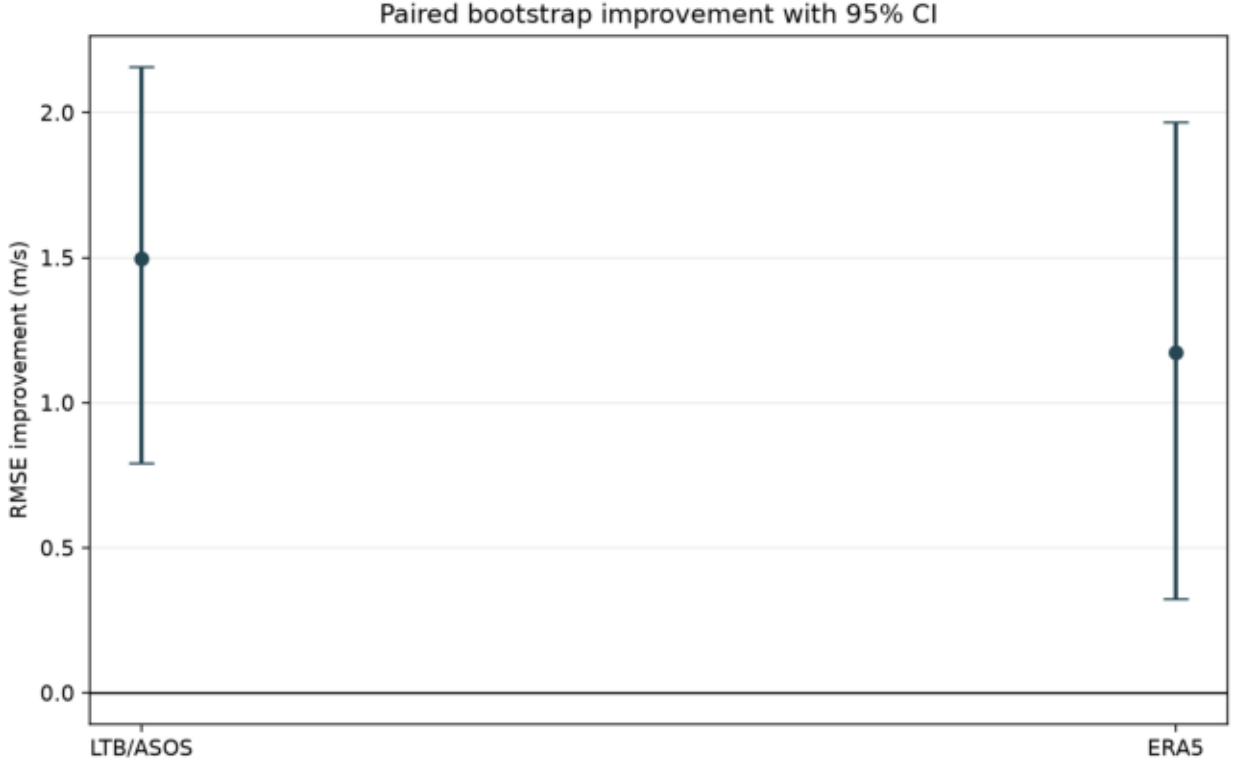
LTB hedefinde ortalama RMSE iyilesmesi: 52.07%\nERA5 hedefinde ortalama RMSE iyile

Sinir: Sonuclar Ocak 2021 ve 18 bagimsiz test ornegine dayanan pilot calismadir.

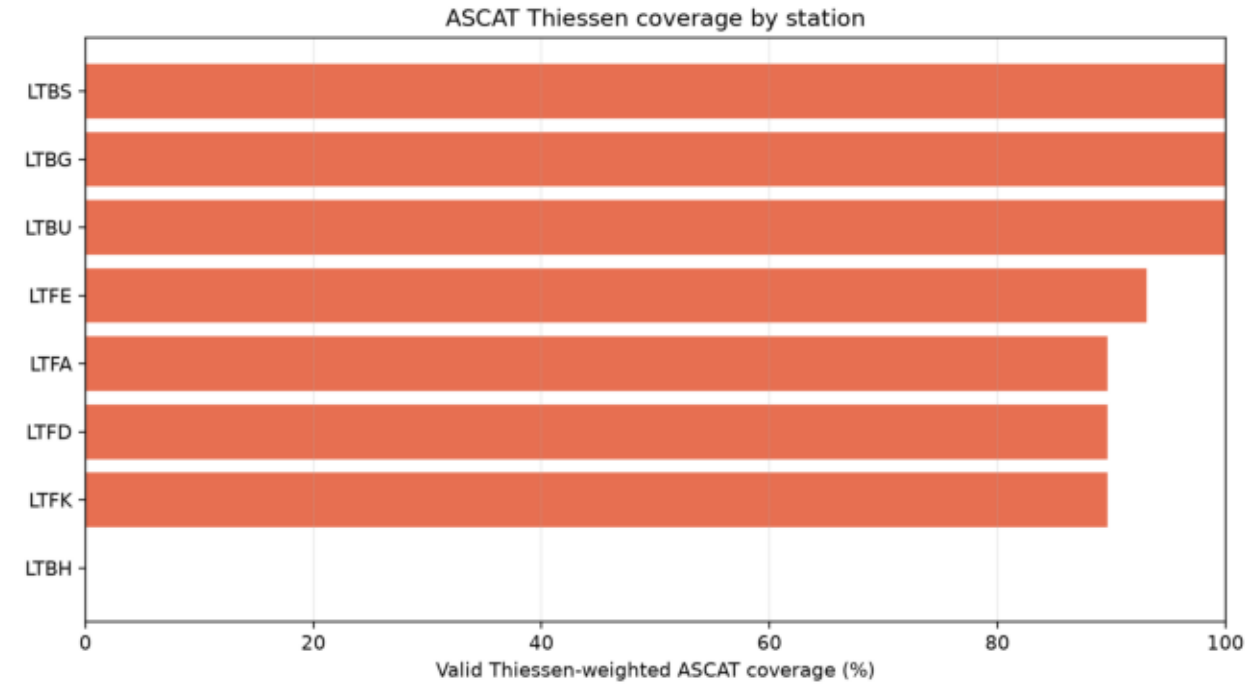
Model Rmse Comparison



Rmse Bootstrap Improvement



Thiessen Coverage



Rolling-origin sonuclar

target	model	samples	mae	rmse	r2
ERA5	TIGGE	18	2.032	2.386	-17.259
ERA5	ASCAT Thiessen XGBoost	18	0.978	1.192	-3.29
ERA5	linear regression	18	1.106	1.202	-4.072
ERA5	random forest	18	1.093	1.396	-5.685
ERA5	TIGGE bias correction	18	1.831	2.141	-13.857
LTB	TIGGE	18	2.645	2.988	-103.187
LTB	ASCAT Thiessen XGBoost	18	1.247	1.432	-16.343
LTB	linear regression	18	1.391	1.626	-28.051
LTB	random forest	18	1.398	1.589	-24.956
LTB	TIGGE bias correction	18	2.443	2.709	-82.828

Formulasyonlar

$$U = \sqrt{u_{10}^2 + v_{10}^2}$$

$$e_t = U_t^{ERA5} - U_t^{TIGGE}$$

$$\hat{U}_t^{corr} = U_t^{TIGGE} + \hat{e}_t(X_t)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{U}_i - U_i)^2}$$

$$z_i = (x_i - \bar{x})/s_x$$